

FEATURE OVERVIEW - UPDATED 2026-06-18

FitRaceStudio

Live cardio racing for studios and events

A real-time race experience with athlete registration, role-based administration, station assignment, Edge Node monitoring, race control, and a large-screen leaderboard.



What the System Enables

A complete in-studio race workflow with a clear split between coach operations and technical administration.

Coach Race Operations

Game Admin provides race setup, start, stop, reset, and read-only station status.

Live Race Display

The dashboard presents rankings, race state, station labels, athletes, teams, and progress.

Technical Administration

System Admin owns Edge Node status, telemetry stream discovery, station assignment, updates, and power actions.

Edge Node Setup

Local Edge screens support signal checks and equipment discovery during installation.

Athlete Registration

Signup is mobile-friendly and focused only on station-based registration.

International Events

The UI includes language switching for mixed-language studios and events.

Role-Based Admin Model

The latest admin split keeps daily race operation simple and moves equipment configuration to technical staff.

Game Admin

- Race type and target setup
- Start, stop, and reset race
- Read-only station status
- Designed for coaches and floor staff

System Admin

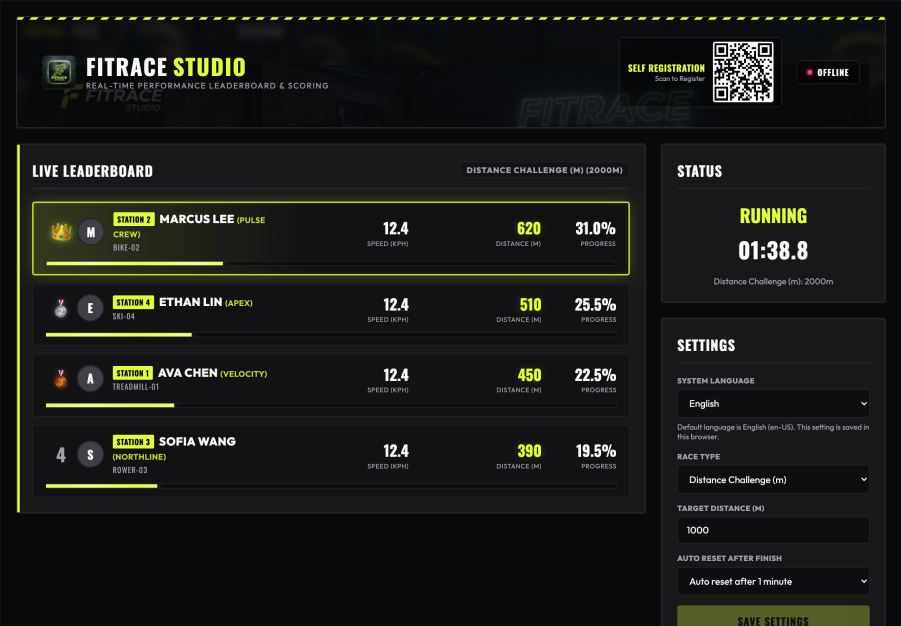
- Edge Node online/offline status
- Telemetry stream discovery
- Station assignment and signup links
- Updates and power controls

Signup

- Athlete name and team entry
- Avatar selection or upload
- Station-based registration link
- No race or system controls

Live Race Dashboard

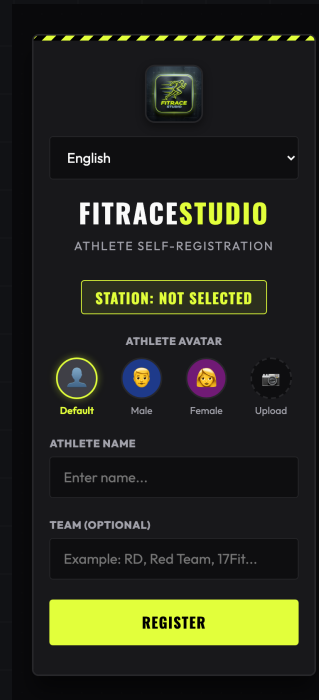
The large-screen view for athletes, coaches, and spectators.



Live leaderboard with race state, ranking, station labels, athlete names, teams, and progress.

Athlete Self-Registration

A mobile-first page for fast station-based signup. Management controls are intentionally kept out of this screen.



The image shows a mobile app interface for athlete self-registration. At the top, there is a yellow and black striped border. Below it is a small logo with the word "FITRACE" and a stylized figure. A language dropdown menu is set to "English". The main title "FITRACESTUDIO" is in large, bold, yellow letters, with "ATHLETE SELF-REGISTRATION" in smaller white letters below it. A yellow button with the text "STATION: NOT SELECTED" is centered. Below this is the "ATHLETE AVATAR" section, which includes four circular icons: "Default" (a person silhouette), "Male" (a male face), "Female" (a female face), and "Upload" (a camera icon). Below the avatars is the "ATHLETE NAME" section with a text input field containing the placeholder "Enter name...". Below that is the "TEAM (OPTIONAL)" section with a text input field containing the placeholder "Example: RD, Red Team, T7Fit...". At the bottom is a large yellow button with the text "REGISTER".

Phone-friendly registration with name, team, avatar, and station context.

Edge Node Setup

Local setup view for signal checks and equipment discovery before technical mapping in System Admin.

FitRace Edge Node

Local Edge Node setup

English

Local web setup :8001

Wi-Fi Signal

Disconnected

Confirm the Edge Node is connected to the AP

wlan0

SSID

Not connected

No Wi-Fi RSSI available. Confirm the Edge Node is connected to fitRace26.

BLE Scan

ADAPTER

hci1 USB dongle

TIMEOUT SECONDS

5

INCLUDE NON-FTMS DEVICES

Scan FTMS Devices

Status

Idle

Elapsed

0.0s

Remaining

0.0s

Ready to scan nearby FTMS equipment.

Discovered Devices

No scan results yet.

Edge setup screen for Wi-Fi status and FTMS device scanning.

Feature Summary

The core functions available across the system screens.

Live Leaderboard

Ranking, race state, station labels, athlete names, teams,

Edge Node Monitoring

Review online/offline status, software version, endpoint, last heartbeat, and equipment streams.

Game Admin Race Control

Select race type, set targets, start, stop, and reset races without exposing device maintenance controls.

Athlete Registration

Phone-friendly signup with name, team, avatar, and station context.

System Admin Station Assignment

Bind equipment streams to station numbers, review athletes, and copy station

System Maintenance

Check updates, download artifacts, stage Hub updates, restart service, reboot, or shut down with confirmations.

Typical Event Flow

From technical setup to live race display.

1 System Setup Technical staff check Edge Nodes and equipment streams in System Admin.	2 Assign Stations Telemetry streams are mapped to numbered physical stations.	3 Register Athletes Participants register by station from phone-friendly signup links.
4 Configure Race Coaches choose the race type and target in Game Admin.	5 Start Live Race The dashboard updates ranking, progress, and timing in real time.	6 Review Results Final leaderboard supports wrap-up, awards, and event moments.

Built for live fitness competitions, studio classes, and event operations.